

T.C.

KONYA TEKNİK ÜNİVERSİTESİ

MÜHENDİSLİK VE DOĞA BİLİMLERİ FAKÜLTESİ

ELEKTRİK–ELEKTRONİK MÜHENDİSLİĞİ BÖLÜMÜ

ROBOTİĞE GİRİŞ ÖDEV 3-2

ROS PID CONTROLLER

Hazırlayan

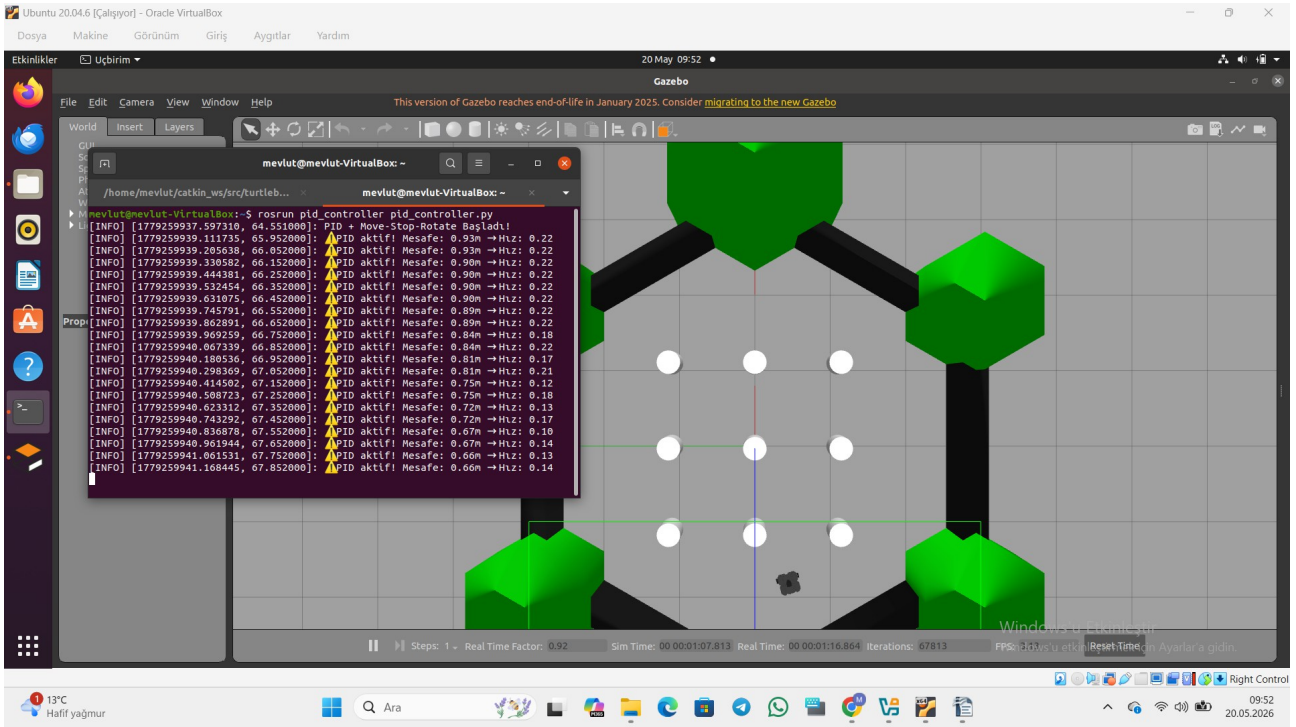
MEVLÜT ÇELİK

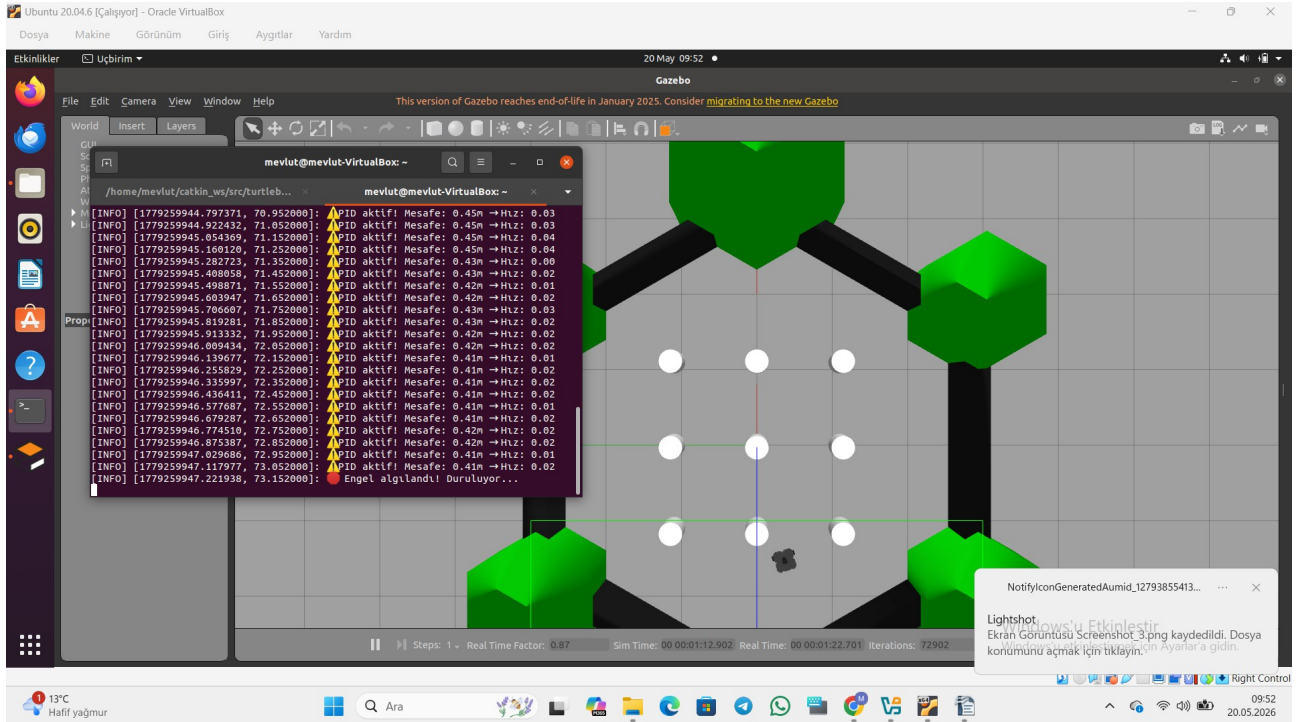
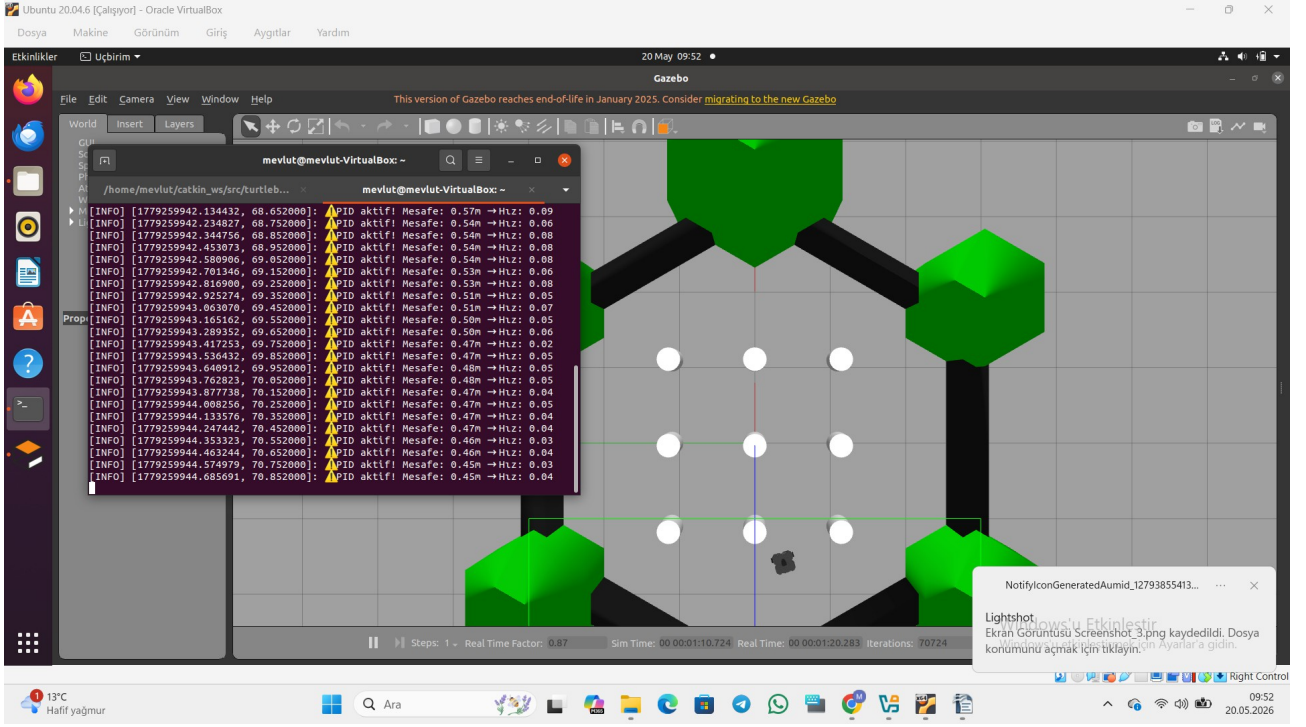
241222006

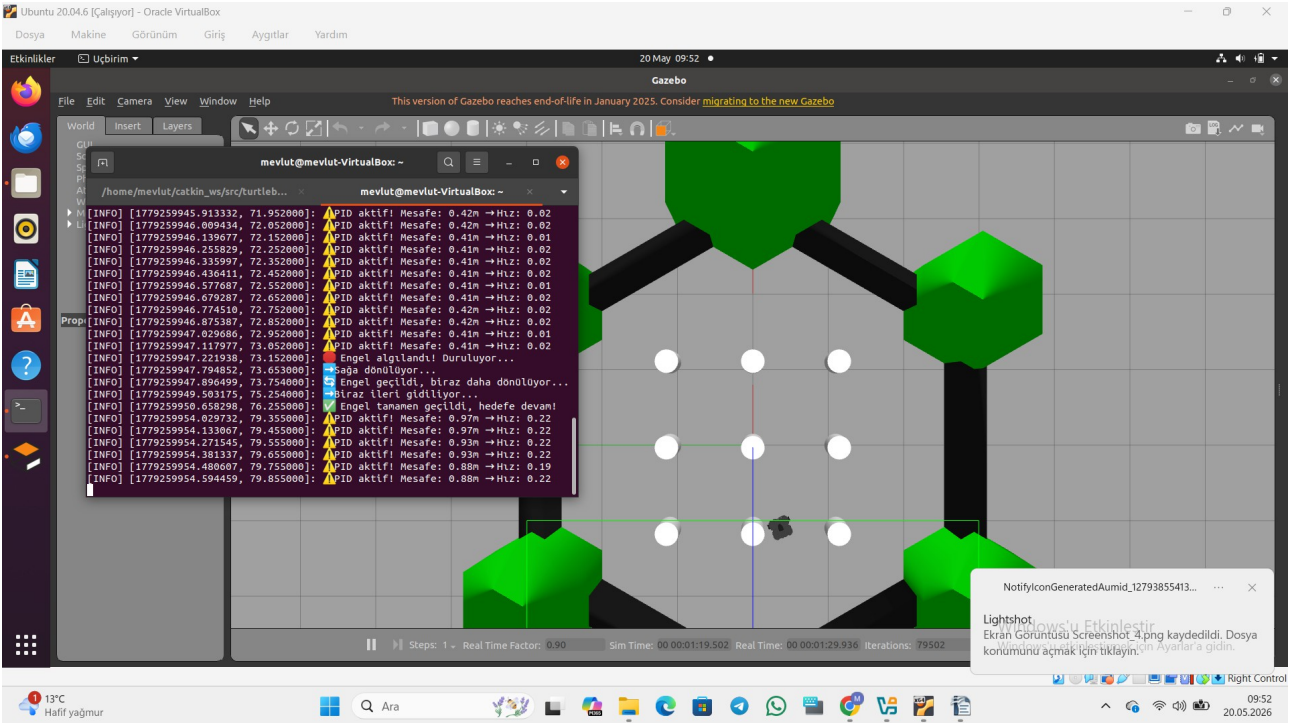
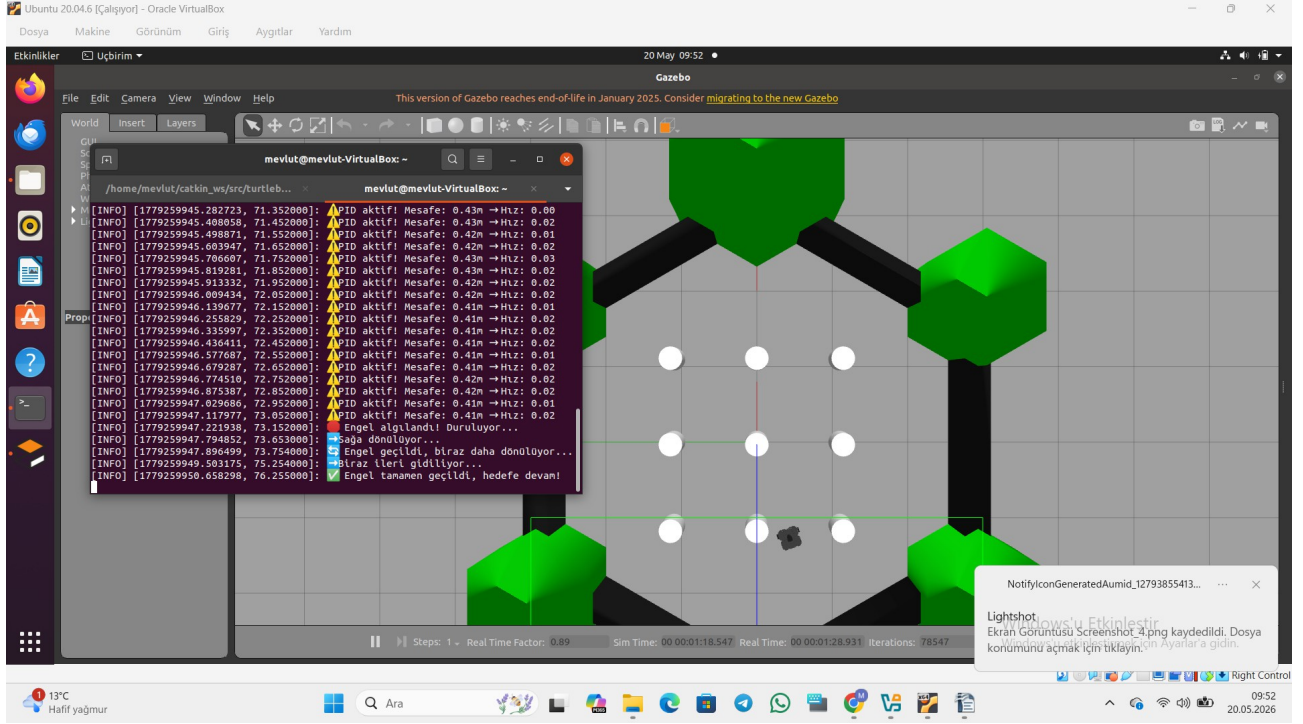
Konya – 2026

```
/home/mevlut/catkin_ws/src/turtlebot3_simulations/turtle...  
mevlut@mevlut-VirtualBox: ~  
mevlut@mevlut-VirtualBox:~$ roslaunch turtlebot3_gazebo turtlebot3_world.launch  
... logging to /home/mevlut/.ros/log/41ec0130-5418-11f1-b45f-4d7ce72ebb60/roslau  
nch-mevlut-VirtualBox-1875.log
```

```
mevlut@mevlut-VirtualBox: ~  
mevlut@mevlut-VirtualBox: ~  
mevlut@mevlut-VirtualBox:~$ rosrn pid_controller pid_controller.py
```








```
mevlut@mevlut-VirtualBox: ~  
/home/mevlut/catkin_ws/src/turtleb... x mevlut@mevlut-VirtualBox: ~  
mevlut@mevlut-VirtualBox:~$ rosrund pid_controller pid_controller.py  
INFO] [1779259937.597310, 64.551000]: PID + Move-Stop-Rotate Başladı!  
INFO] [1779259939.111735, 65.952000]: PID aktif! Mesafe: 0.93m → Hız: 0.22  
INFO] [1779259939.205638, 66.052000]: PID aktif! Mesafe: 0.93m → Hız: 0.22  
INFO] [1779259939.330582, 66.152000]: PID aktif! Mesafe: 0.90m → Hız: 0.22  
INFO] [1779259939.444381, 66.252000]: PID aktif! Mesafe: 0.90m → Hız: 0.22  
INFO] [1779259939.532454, 66.352000]: PID aktif! Mesafe: 0.90m → Hız: 0.22  
INFO] [1779259939.631075, 66.452000]: PID aktif! Mesafe: 0.90m → Hız: 0.22  
INFO] [1779259939.745791, 66.552000]: PID aktif! Mesafe: 0.89m → Hız: 0.22  
INFO] [1779259939.862891, 66.652000]: PID aktif! Mesafe: 0.89m → Hız: 0.22  
INFO] [1779259939.969259, 66.752000]: PID aktif! Mesafe: 0.84m → Hız: 0.18  
INFO] [1779259940.067339, 66.852000]: PID aktif! Mesafe: 0.84m → Hız: 0.22  
INFO] [1779259940.180536, 66.952000]: PID aktif! Mesafe: 0.81m → Hız: 0.17  
INFO] [1779259940.298369, 67.052000]: PID aktif! Mesafe: 0.81m → Hız: 0.21  
INFO] [1779259940.414502, 67.152000]: PID aktif! Mesafe: 0.75m → Hız: 0.12  
INFO] [1779259940.508723, 67.252000]: PID aktif! Mesafe: 0.75m → Hız: 0.18  
INFO] [1779259940.623312, 67.352000]: PID aktif! Mesafe: 0.72m → Hız: 0.13  
INFO] [1779259940.743292, 67.452000]: PID aktif! Mesafe: 0.72m → Hız: 0.17  
INFO] [1779259940.836878, 67.552000]: PID aktif! Mesafe: 0.67m → Hız: 0.10  
INFO] [1779259940.961944, 67.652000]: PID aktif! Mesafe: 0.67m → Hız: 0.14  
INFO] [1779259941.061531, 67.752000]: PID aktif! Mesafe: 0.66m → Hız: 0.13  
INFO] [1779259941.168445, 67.852000]: PID aktif! Mesafe: 0.66m → Hız: 0.14  
INFO] [1779259941.275112, 67.952000]: PID aktif! Mesafe: 0.63m → Hız: 0.09  
INFO] [1779259941.372782, 68.052000]: PID aktif! Mesafe: 0.63m → Hız: 0.12
```

```
mevlut@mevlut-VirtualBox: ~  
/home/mevlut/catkin_ws/src/turtleb... x mevlut@mevlut-VirtualBox: ~  
[INFO] [1779259941.168445, 67.852000]: PID aktif! Mesafe: 0.66m → Hız: 0.14  
[INFO] [1779259941.275112, 67.952000]: PID aktif! Mesafe: 0.63m → Hız: 0.09  
[INFO] [1779259941.372782, 68.052000]: PID aktif! Mesafe: 0.63m → Hız: 0.12  
[INFO] [1779259941.507461, 68.152000]: PID aktif! Mesafe: 0.61m → Hız: 0.09  
[INFO] [1779259941.628325, 68.252000]: PID aktif! Mesafe: 0.61m → Hız: 0.11  
[INFO] [1779259941.777738, 68.352000]: PID aktif! Mesafe: 0.58m → Hız: 0.07  
[INFO] [1779259941.918535, 68.452000]: PID aktif! Mesafe: 0.58m → Hız: 0.10  
[INFO] [1779259942.021724, 68.552000]: PID aktif! Mesafe: 0.57m → Hız: 0.08  
[INFO] [1779259942.134432, 68.652000]: PID aktif! Mesafe: 0.57m → Hız: 0.09  
[INFO] [1779259942.234827, 68.752000]: PID aktif! Mesafe: 0.54m → Hız: 0.06  
[INFO] [1779259942.344756, 68.852000]: PID aktif! Mesafe: 0.54m → Hız: 0.08  
[INFO] [1779259942.453073, 68.952000]: PID aktif! Mesafe: 0.54m → Hız: 0.08  
[INFO] [1779259942.580906, 69.052000]: PID aktif! Mesafe: 0.54m → Hız: 0.08  
[INFO] [1779259942.701346, 69.152000]: PID aktif! Mesafe: 0.53m → Hız: 0.06  
[INFO] [1779259942.816900, 69.252000]: PID aktif! Mesafe: 0.53m → Hız: 0.08  
[INFO] [1779259942.925274, 69.352000]: PID aktif! Mesafe: 0.51m → Hız: 0.05  
[INFO] [1779259943.063070, 69.452000]: PID aktif! Mesafe: 0.51m → Hız: 0.07  
[INFO] [1779259943.165162, 69.552000]: PID aktif! Mesafe: 0.50m → Hız: 0.05  
[INFO] [1779259943.289352, 69.652000]: PID aktif! Mesafe: 0.50m → Hız: 0.06  
[INFO] [1779259943.417253, 69.752000]: PID aktif! Mesafe: 0.47m → Hız: 0.02  
[INFO] [1779259943.536432, 69.852000]: PID aktif! Mesafe: 0.47m → Hız: 0.05  
[INFO] [1779259943.640912, 69.952000]: PID aktif! Mesafe: 0.48m → Hız: 0.05  
[INFO] [1779259943.762823, 70.052000]: PID aktif! Mesafe: 0.48m → Hız: 0.05  
[INFO] [1779259943.877738, 70.152000]: PID aktif! Mesafe: 0.47m → Hız: 0.04
```

```
mevlut@mevlut-VirtualBox: ~  
/home/mevlut/catkin_ws/src/turtleb... mevlut@mevlut-VirtualBox: ~  
INFO] [1779259945.408058, 71.452000]: ⚠ PID aktif! Mesafe: 0.43m → Hız: 0.02  
INFO] [1779259945.498871, 71.552000]: ⚠ PID aktif! Mesafe: 0.42m → Hız: 0.01  
INFO] [1779259945.603947, 71.652000]: ⚠ PID aktif! Mesafe: 0.42m → Hız: 0.02  
INFO] [1779259945.706607, 71.752000]: ⚠ PID aktif! Mesafe: 0.43m → Hız: 0.03  
INFO] [1779259945.819281, 71.852000]: ⚠ PID aktif! Mesafe: 0.43m → Hız: 0.02  
INFO] [1779259945.913332, 71.952000]: ⚠ PID aktif! Mesafe: 0.42m → Hız: 0.02  
INFO] [1779259946.009434, 72.052000]: ⚠ PID aktif! Mesafe: 0.42m → Hız: 0.02  
INFO] [1779259946.139677, 72.152000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.01  
INFO] [1779259946.255829, 72.252000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.02  
INFO] [1779259946.335997, 72.352000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.02  
INFO] [1779259946.436411, 72.452000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.02  
INFO] [1779259946.577687, 72.552000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.01  
INFO] [1779259946.679287, 72.652000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.02  
INFO] [1779259946.774510, 72.752000]: ⚠ PID aktif! Mesafe: 0.42m → Hız: 0.02  
INFO] [1779259946.875387, 72.852000]: ⚠ PID aktif! Mesafe: 0.42m → Hız: 0.02  
INFO] [1779259947.029686, 72.952000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.01  
INFO] [1779259947.117977, 73.052000]: ⚠ PID aktif! Mesafe: 0.41m → Hız: 0.02  
INFO] [1779259947.221938, 73.152000]: ⚠ Eng el algılandı! Duruluyor...  
INFO] [1779259947.794852, 73.653000]: ➡ Sağa dönölüyor...  
INFO] [1779259947.896499, 73.754000]: ➡ Eng el geçildi, biraz daha dönölüyor...  
INFO] [1779259949.503175, 75.254000]: ➡ Biraz ileri gidiliyor...  
INFO] [1779259950.658298, 76.255000]: ✓ Eng el tamamen geçildi, hedefe devam!  
INFO] [1779259954.029732, 79.355000]: ⚠ PID aktif! Mesafe: 0.97m → Hız: 0.22  
INFO] [1779259954.133067, 79.455000]: ⚠ PID aktif! Mesafe: 0.97m → Hız: 0.22
```

